inflateR

Historical currency adjustment in R

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The question

What is an amount in one year's money worth in another year's money?

Thirteen currencies, two inflation measures, one tidy interface.

Why it matters

- **Journalism**: what is that 1970 salary worth in today's money?
- **Economic history**: deflating nineteenth- and twentieth-century wages to a common unit
- **Accounting and compliance**: revaluing historical assets, legacy contracts, and long-dated liabilities
- **Public policy**: comparing government spending, wages, and benefits across decades

What is already out there

- **`priceR`**: pricing-related toolkit with some inflation helpers, broader scope[1]
- **`quantmod`**: pulls CPI series from FRED but leaves adjustment to the user
- **`WDI`**: fetches World Bank series but does not adjust values directly
- **Bespoke spreadsheets**: ad hoc, single-currency, hard to audit

The gap: **no purpose-built tool that bundles the canonical indices and exposes a symmetric pair of adjustment functions**.

[1] Condamin (2022), *priceR: Economics and Pricing Tools*, R package on CRAN.

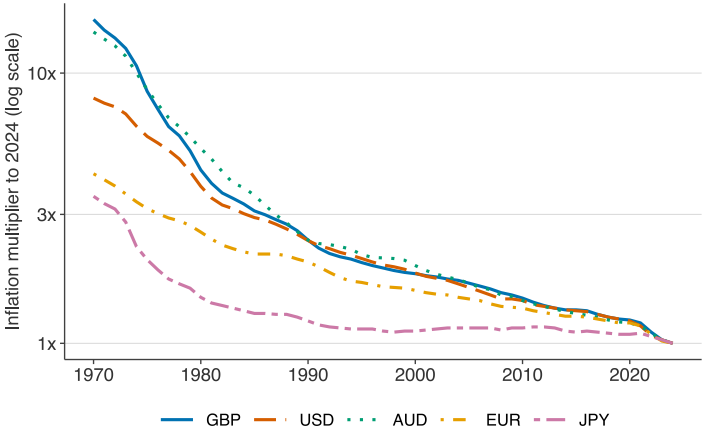
What inflateR offers

1. **Coverage**: thirteen currencies (GBP, USD, AUD, EUR, CAD, JPY, CNY, CHF, NZD, INR, KRW, BRL, NOK), 1960 to 2024
2. **Interface**: four adjustment functions in two symmetric pairs, one cumulative/annualised rate helper, one metadata helper
3. **Provenance**: zero runtime dependencies, no API key, no network access, World Bank WDI data bundled

On CRAN since March 2026. CPI and GDP deflator both exposed[2].

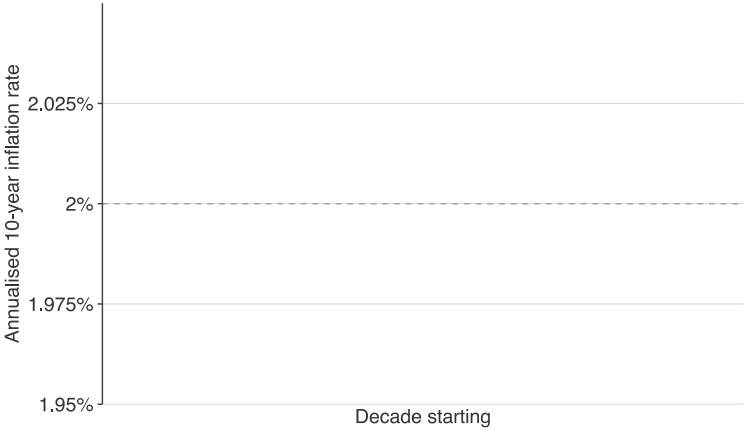
[2] World Bank Development Indicators, series `FP.CPI.TOTL` and `NY.GDP.DEFL.ZS`, rescaled to 2020 = 100.

Multipliers: GBP, USD, AUD, EUR, JPY to 2024



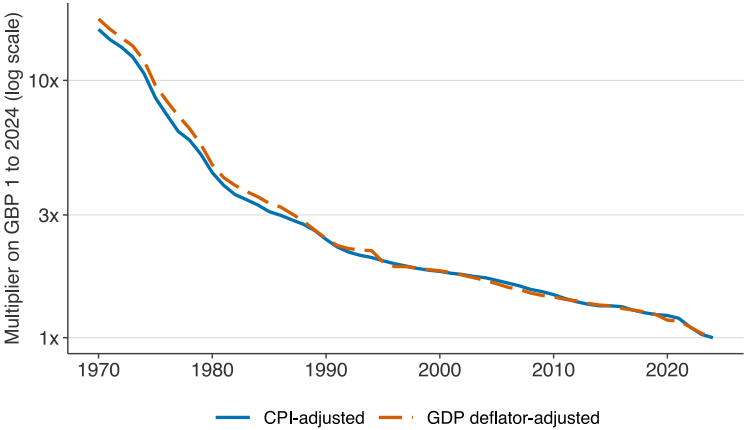
Multiplier to convert one unit of historical currency into 2024 money, five currencies, 1960 to 2024. Exports: `adjust_inflation()`, `adjust_real()`.

Rolling 10-year annualised inflation



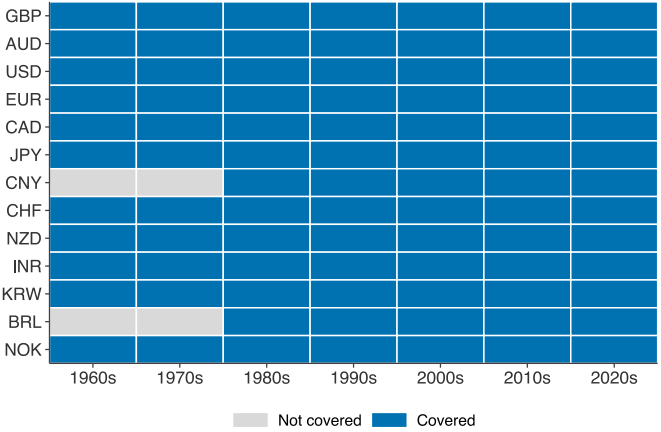
Rolling 10-year annualised CPI inflation, five currencies, 1970 to 2024. Exports: `inflation_rate()`.

CPI versus GDP deflator for GBP



Two measures of GBP inflation, 1970 to 2024, rebased to 2020 = 100. Exports: `adjust_inflation()`, `adjust_real()`.

Coverage: thirteen currencies, six decades



Data availability heatmap, thirteen currencies, 1960s to 2020s. Exports: `list_currencies()`.

Central formulas

Forward adjustment (historical to present):

$$\text{adjust_inflation}(a,\ s \to t) = a \cdot \frac{I_t}{I_s} \tag{1}$$

Cumulative inflation rate between years s and t:

$$\pi_{s \to t} = \frac{I_t}{I_s} - 1 \tag{2}$$

Annualised inflation rate for $n = t - s$ years:

$$\pi_{\text{ann}} = \left(\frac{I_t}{I_s}\right)^{1/n} - 1 \tag{3}$$

Package at a glance

Function families:

- **CPI adjustment**: `adjust_inflation()`, `historical_value()`
- **GDP deflator adjustment**: `adjust_real()`, `historical_real()`
- **Rates**: `inflation_rate()` (cumulative or annualised)
- **Metadata**: `list_currencies()`

Deps: base R only. R $\geq$ 3.5.0. No API key, no network.

Direction-measure matrix

Two directions (forward and reverse) times two measures (CPI and deflator) = four core functions. Every function accepts ISO code or country name.

Minimal working example

```r
library(inflateR)

# Forward: 1990 GBP 30,000 salary in today's money
adjust_inflation(30000, from_year = 1990, currency = "GBP")

# Reverse: today's GBP 50,000 in 1990 equivalent
historical_value(50000, to_year = 1990, currency = "GBP")

# Rate: annualised CPI inflation for GBP, 1970 to 2024
inflation_rate("GBP", from_year = 1970, to_year = 2024,
               annualised = TRUE)
```

Every function is vectorised over `amount`, so bulk conversion of a column of historical prices requires no loop.

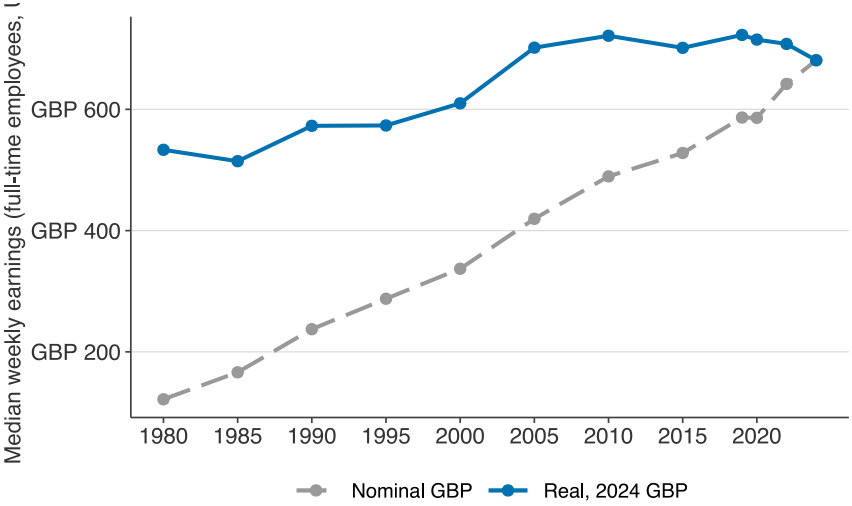
UK real median weekly earnings, 1980 to 2024

Data. ONS Annual Survey of Hours and Earnings (ASHE). Median gross weekly pay for full-time employees, 1980 to 2024 as published.

Question. *Have UK real wages stagnated since the 2010 peak?*

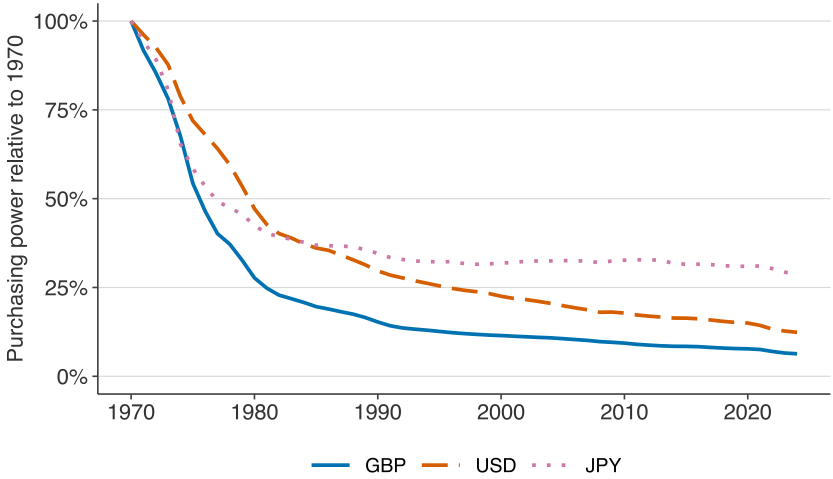
Why this case. Canonical UK applied-labour question over the past fifteen years[3]. Nominal series hides the phenomenon entirely; CPI deflation via a single function call reveals it. Reproducible with one line of `adjust_inflation()`.

[3] Blundell, Joyce, Keiller & Ziliak (2023), *The dynamics of real wages in the UK*, and Giupponi & Machin (2023) on wage stagnation.



UK median weekly earnings, nominal and real (2024 GBP), 1980 to 2024. Grey dashed: nominal GBP as published. Blue solid: real GBP deflated by `adjust_inflation()` using World Bank CPI. Real earnings rose from GBP 533 in 1980 to a peak of GBP 721 in 2010 (+35 per cent), then fell to GBP 681 by 2024 (−5.6 per cent post-2010). Source: ONS ASHE.

Purchasing power has collapsed for GBP, USD, and JPY



Purchasing power of 1 unit of currency relative to 1970. 1970 to 2024. Computed via `1 / adjust_inflation(1`

What inflateR does not yet do

- **World Bank WDI only**: native methodologies (ONS RPI, BLS chained CPI) differ modestly
- **No PPP conversion**: cross-country comparisons at a point in time need a separate step
- **Annual data only**: no monthly or quarterly adjustment out of the box
- **1960 start**: pre-1960 history requires users to supply an alternative series

v0.2.0 roadmap: expanded currency coverage, optional PPP-adjusted cross-country conversion, higher-frequency data, and an `inflation_chart()` helper for plot output.

Contact, code, paper

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`charlescoverdale.github.io/publications`

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